

COMSATS University Islamabad, Pakistan

Project Name

A project presented to

COMSATS Institute of Information Technology, Islamabad

In partial fulfillment

of the requirement for the degree of

Bachelor of Science in Computer Science (20xx-20xx)

By

Student Name 1 CIIT/SP09-BCS-xxx/ISB

Student Name 2 CIIT/SP09-BCS-xxx/ISB

DECLARATION

We hereby declare that this software, neither whole nor as a part has been copied out from any source. It is further declared that we have developed this software and accompanied report entirely on the basis of our personal efforts. If any part of this project is proved to be copied out from any source or found to be reproduction of some other. We will stand by the consequences. No Portion of the work presented has been submitted of any application for any other degree or qualification of this or any other university or institute of learning.

Student Name 1

Student Name 2

CERTIFICATE OF APPROVAL

It is to certify that the final year project of BS (CS) “Project title” was developed by **STUDENT 1 NAME (CIIT/FAXX-BCS/SE/TN-000)** and **STUDENT 2 NAME (CIIT/FAXX-BCS/SE/TN-000)** under the supervision of “SUPERVISOR NAME” and co supervisor “CO-SUPERVISOR NAME” and that in (their/his/her) opinion; it is fully adequate, in scope and quality for the degree of Bachelors of Science in Computer Sciences.

Supervisor

External Examiner

Head of Department

(Department of Computer Science)

Executive Summary

In public places, there is often a need for monitoring people and different activities going on, which can be referred later for many reasons including security. Appointing humans for this task involves many problems such as increased employee hiring, accuracy problem, trust, no proof for later use, and also the fact that a human can remember things till a certain time limit. Talking about the current security system, they use dumb still cameras with a continuous recording facility ir-respective of the fact that any event may happen or not. Moreover they are usually pointing at a specific user defined locations so more than one cameras are required to cover the entire region.

To prevent all these problems from prevailing, the CSCS is developed. It is a surveillance system, which provides solution to many of these problems. It is a stand-alone application which doesn‟t require any computer to operate. It monitors different situations using a camera which is able to rotate intelligently based on sensor messages and captures the scene in the form of video or photos later reference as well.

Customizable **S**urveillance **C**ontrol **S**ystem **(CSCS)** is a surveillance system that can be assigned a sensor type as in our case a heat sensor is used, it works accordingly, rotates the camera upon event detection and perform user defined actions like capturing video and stores them, for the future use.

It is an embedded system consisting of Linux fox kit with embedded a running server application also a camera, USB storage device and a sensor node base station is attached with fox kit. LAN communication is used by user to download the videos and to operate the system manually.

Acknowledgement

All praise is to Almighty Allah who bestowed upon us a minute portion of His boundless knowledge by virtue of which we were able to accomplish this challenging task.

We are greatly indebted to our project supervisor “Dr. Majid Iqbal Khan” and our Co-Supervisor “Mr. Mukhtar Azeem”. Without their personal supervision, advice and valuable guidance, completion of this project would have been doubtful. We are deeply indebted to them for their encouragement and continual help during this work.

And we are also thankful to our parents and family who have been a constant source of encouragement for us and brought us the values of honesty & hard work.

Student Name 1 Student Name 2

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Abbreviations

SRS	Software Require Specification
PC	Personal Computer

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Introduction

This chapter provides the overview of the project. The first paragraph of every chapter should provide the chapter summary.

Brief

A very brief introduction of project work, outcome of your work, tools, methodology used & highlights of discussions in various chapters of report.

Relevance to Course Modules

A brief explanation of how your project is related to various courses studied during BCS.

Project Background

It includes explanation of the idea behind the project. For example if the project is related to VoIP then this section describes that what is voice over IP & how it works.

Literature Review

This section describes current trends/ research/ products etc. related to your project.

Analysis from Literature Review (in the context of your project)

This section provides an analytical discussion of your work in comparison with discussion in literature review.

Methodology and Software Lifecycle for This Project

A brief discussion of methodology and SDLC model selected for this project.

Rationale behind Selected Methodology

Why you selected above methodology (such as structural and Object Oriented) and software life cycle for this project?

Rationale behind Selected Methodology

It is the example of third and last level heading. Please do not insert further levels in numbers. Use different format style e.g. italic to highlight the important text.

Problem Definition

This chapter discusses the precise problem to be solved. It should extend to include the outcome.

Problem Statement

Problem statement goes here.

Deliverables and Development Requirements

Deliverables and development requirements.

Current System (if applicable to your project)

A brief description of an existing system.

The following figure is a sample figure, Figure 2.1. You are required to follow the same style of numbering and caption for the whole report.

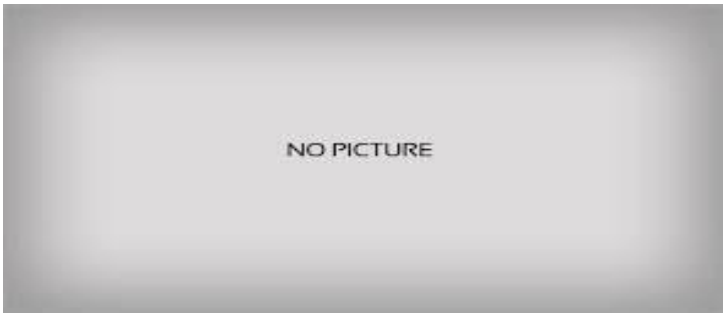


Figure 2.1: Sample picture

The following table (Table 2.1) is sample table; you are required to follow the same style of numbering and caption for the whole report.

Table 32.1: Sample Table

Header 1	**Header 2**	**Header 3**
Text	Text	Text

The following list style is the sample to consistently follow in the whole report.

- List items 1
- List items 2

Requirement Analysis

The following parts of Software Requirements Specification (SRS) report should be included

in this chapter.

Use Cases Diagram(s)

Detailed Use Case

Functional Requirements

Non-Functional Requirements

Design and Architecture

The following parts of Software Design Description (SDD) report should be included in this chapter.

System Architecture

Data Representation [Diagram + Description]

Process Flow/Representation

Design Models [along with descriptions]

Implementation

This chapter will discuss implementation details supported by UML diagrams (if applicable). You will not put your source code here. Any of the following sections may be included based on your project.

Algorithm

Mention the algorithm(s) used in your project to get the work done with regards to major modules. Provide a pseudocode **OR** a natural language explanation regarding the functioning of main features. Be sure to use the correct syntax and semantics for algorithm representations.

External APIs

Describe the APIs used in the following table.

Table 8 Details of APIs used in the project

Name of API	**Description of API**	**Purpose of usage**	**List down the function/class name in which it is used**

User Interface

Details about user interface with descriptions.

Testing and Evaluation

This chapter may include the following sections. (Students are required to perform the testing both manually and automatedly).

Manual Testing

System testing

Once the system has been successfully developed, testing has to be performed to ensure that the system working as intended. This is also to check that the system meets the requirements stated earlier. Besides that, system testing will help in finding the errors that may be hidden from the user. There are few types of testing which includes the unit testing, functional testing and integration testing. The testing must be completed before it is being deploy for user to use.

Unit Testing

Unit Testing 1: Login as FYP Committee
Testing Objective: To ensure the login form is working correctly

No.	Test case/Test script	Attribute and value	Expected result	Result
1.	Verify user login after click on the „Login“ button on login form with correct input data	Username: L001 Password: 1234	Successfully log into the main page of the system as FYP Committee member.	Pass
2.				

Unit Testing 2: Edit Profile
Testing Objective: To ensure the edit profile form is working properly.

No.	Test case/Test script	Attribute and value	Expected result	Result
1.				
2.				

Functional Testing

The functional testing will take place after the unit testing. In this functional testing, the functionality of each of the module is tested. This is to ensure that the system produced meets the specifications and requirements.

Functional Testing 1: Login with different roles
Objective: To ensure that the correct page with the correct navigation bar is loaded.

No.	Test case/Test script	Attribute and value	Expected result	Result
1.	Login as a „FYP Committee‟ member.	Username: L001 Password: 1234	Main page for the FYP Committee member is loaded with the FYP Committee navigation bar	Pass
2.				

Integration Testing

No.	Test case/Test script	Attribute and value	Expected result	Result
1.	Login as “FYP Committee” member	Username: L001 Password: 1234	Login successful and the FYP Committee page with its navigation bar is loaded and in the view profile page	Pass
2.	Upload student record for Project 1	-	File successfully uploaded and return to the upload page. Student records are updated.	Pass
3.	View supervising student	-	The list of supervisees shown on the screen.	Pass

Automated Testing:

Tools used:

Tool Name	Tool Description	Applied on [list of related test cases / FR / NFR]	Results

Conclusion and Future Work

This chapter concludes the project and highlights future work.

Conclusion

Future Work

References

References to any book, journal paper or website should properly be acknowledged. Please consistently follow the style. The following are few examples of different resources i.e. journal article, book, and website.

1. Lyda M.S. Lau, Jayne Curson, Richard Drew, Peter Dew and Christine Leigh, (1999), Use Of VSP Resource Rooms to Support Group Work in a Learning Environment, ACM 99, pp-2. **(Journal paper example)**

2. Hideyuki Nakanishi, Chikara Yoshida, Toshikazu Nishmora and TuruIshada, (1996), FreeWalk: Supporting Casual Meetings in a Network, pp 308-314 **(paper on web)** http://www.acm.org/pubs/articles/proceedings/cscw/240080/p308-nakanishi.pdf

3. Ali Behforooz& Frederick J.Hudson, (1996), Software Engineering Fundamentals, Oxford University Press. Chapter 8, pp255-235. **(book reference example)**

4. Page Author, Page Title, http://www.bt.com/bttj/archive.htm, Last date accessed**. (web site)**